

OCI Link Budget — Results Summary

GEN1 (53.125 GBd NRZ, OCI MSA v1.0) and GEN2 (106.25 GBd NRZ CPO) per-channel OMA-domain budgets at BER 10^{-12} ($Q = 7.035$). Results only — methodology, derivations, and provenance in [OCI_Link_Budget_Report.md](#) and [Methodology_Provenance.md](#). All numbers reproduce from scripts 01–07.

1. Targets

BER	Q	Role
2.4×10^{-4}	3.49	OCI MSA compliance threshold
10^{-12}	7.035	Internal design target (KP4 FEC = pure margin)

2. GEN1 — 53.125 GBd NRZ

Floor –12.93 dBm OMA (measured TIA: $i_n = 3.17\text{ }\mu\text{A rms}$, 29.3 GHz, $R = 0.876\text{ A/W}$).

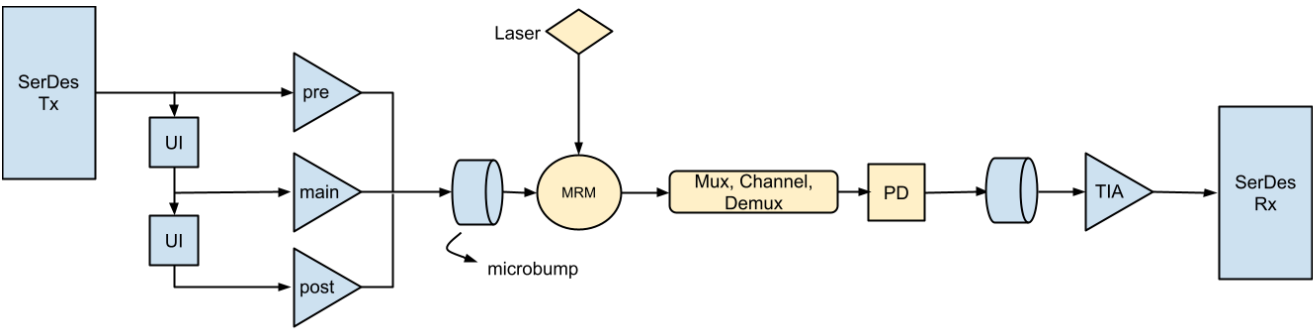
Penalty	dB
ER / shot	0.18
RIN (–138 dB/Hz)	0.68
MPI (–24 dB ends)	0.24
ISI residual (post-DFE)	0.64
CD	0.01
Jitter (TJ 0.241 UI)	0.61
Crosstalk	0.36
Threshold offset	0.21
Dark current	0.00
Stack total	2.93

Required OMA at Rx = –12.93 + 2.93 = **–10.00 dBm**.

Closure scenario	Tx OMA	–IL	–TDEC	At Rx	Margin
Spec-min Tx (TDEC 1.4 or 3.4 dB)	–5.5 / –3.5	2.5	1.4 / 3.4	–9.40	+0.60 dB
Realistic Tx	–3.2	2.5	2.0	–7.70	+2.30 dB

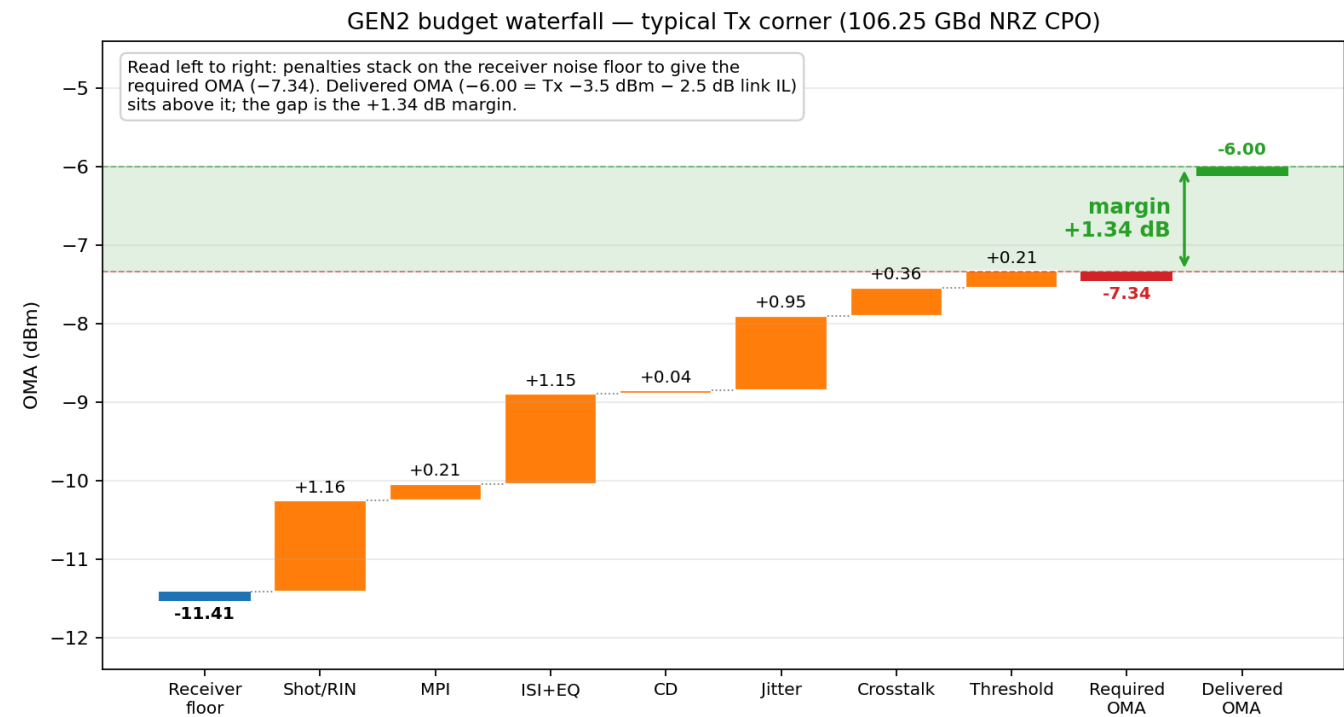
MPI finding: at the spec's –19 dB end reflectance the MPI penalty is 0.51 dB — 2.5× the MSA's 0.2 dB allocation. **≤ –24 dB ends required** (enforced both generations; GEN1 books 0.24 dB, GEN2 0.205 dB).

3. GEN2 — 106.25 GBd NRZ CPO



Rate scaling	GEN1 → GEN2
UI	18.82 → 9.412 ps
Nyquist	26.56 → 53.125 GHz
Noise bandwidth ($1.5 \times f_{3dB}$)	43.9 → 87 GHz
TJ at 10^{-12}	4.5 → 3.30 ps

Target-class TIA floor **-11.41 dBm** (58 GHz Butterworth-2, $i_n = 4.5 \mu\text{A rms}$).



Penalties stack on the receiver floor (-11.41) to give the required OMA (-7.34); the delivered OMA (-6.00) sits above it, and the gap is the +1.34 dB margin.

Penalty	GEN1-CPO @ 53 GBd	GEN2 @ 106 GBd
ER/shot + RIN	0.86	1.16
MPI	0.21	0.21

Penalty	GEN1-CPO @ 53 GBd	GEN2 @ 106 GBd
ISI + EQ net	1.86	1.15
CD	0.01	0.04
Jitter	0.73	0.95
Crosstalk	0.36	0.36
Threshold	0.21	0.21
Stack total	4.23	4.07

Closure at Tx OMA -3.5 dBm (Rx -6.0 dBm)	Floor	Stack	Required	Margin
Fast Tx (0.35 UI)	-11.41	3.74	-7.67	+1.67 dB
Typical Tx (0.45 UI)	-11.41	4.07	-7.34	+1.34 dB
Max Tx (0.60 UI)	-11.41	4.62	-6.79	+0.79 dB

Bottom line: feasible, but specification-sensitive — closes at every modeled corner; the doubled rate moves the problem into the receiver (§4 below).

4. Derived GEN2 TIA requirements

Parameter	Requirement
Bandwidth f_{3dB} (diff.)	50–64 GHz window (53–58 GHz target)
Input-referred noise	$\leq 4.0 \mu\text{A rms}$ over $B_n = 1.5 \times f_{3dB}$ (6.5 μA absolute fail)
Noise density mask	$\leq 14 \text{ pA}/\sqrt{\text{Hz}}$ band avg; $\leq 16 \text{ pA}/\sqrt{\text{Hz}}$ spot (1–53 GHz); measure to $\geq 90 \text{ GHz}$
Magnitude peaking	$\leq 1.0 \text{ dB}$ (DC–53 GHz), monotonic rolloff
Group-delay ripple	$\leq 3 \text{ ps p-p}$ (2–40 GHz)
Transimpedance gain	$\geq 57 \text{ dB}\Omega$ differential at max gain
Overload / dynamic range	160–700 μApp ; AGC $\geq 14 \text{ dB}$ range, $\leq 1 \text{ dB}$ steps
DC cancellation	$\geq 750 \mu\text{A}$
LF cutoff	$\leq 1 \text{ MHz}$ (0.05 dB BLW at 72-bit CID)

5. Device trade study (margins at Tx OMA -3.5 dBm)

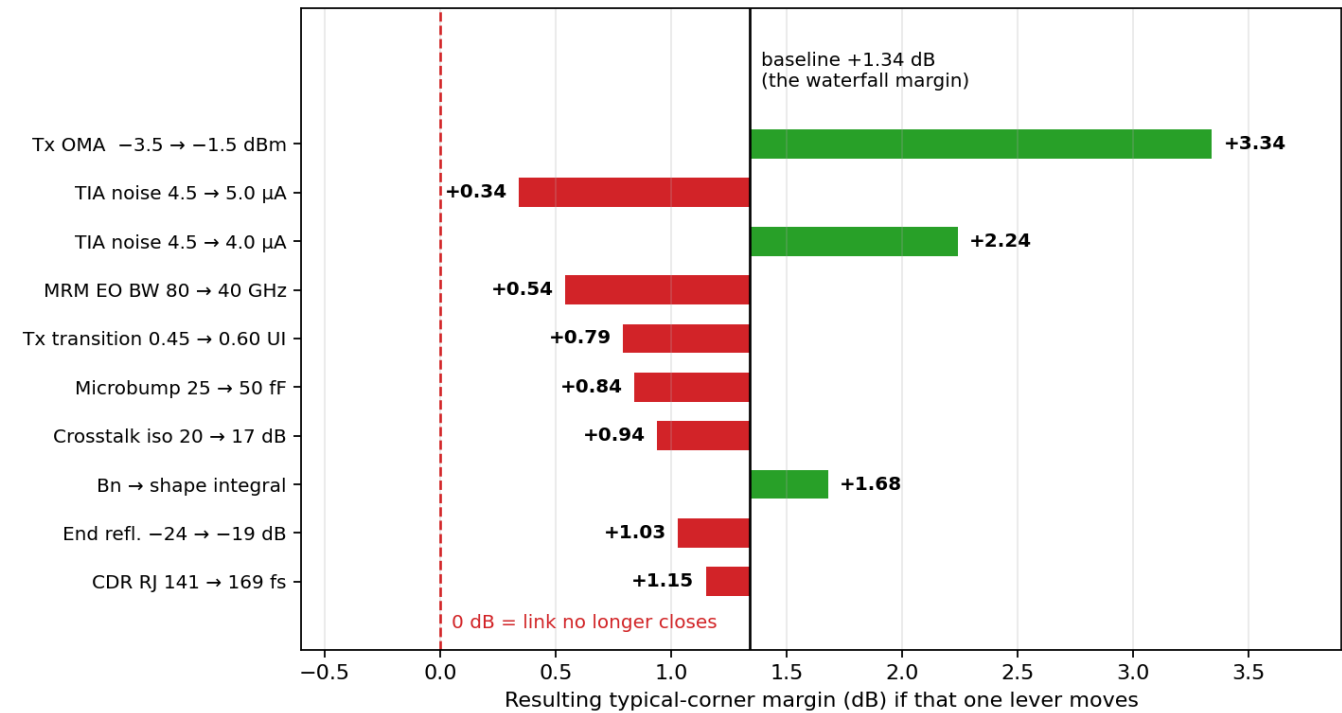
Tx case	A @ 3 $\mu\text{A}/50 \text{ G}$	A @ 4 $\mu\text{A}/50 \text{ G}$	B @ 4 $\mu\text{A}/60 \text{ G}$	B @ 5 $\mu\text{A}/60 \text{ G}$
FIR3, typ driver (0.45 UI)	+2.81	+1.63	+1.98	+1.06
FIR3, slow driver (0.60 UI)	+2.22	+1.04	+1.40	+0.48
no-FIR, drv 60 G + MRM 80 GHz	+2.80	+1.62	+1.83	+0.92

Tx case	A @ 3 μ A/50 G	A @ 4 μ A/50 G	B @ 4 μ A/60 G	B @ 5 μ A/60 G
no-FIR, drv 60 G + MRM 60 GHz	+2.50	+1.32	+1.63	+0.72
no-FIR, drv 60 G + MRM 50 GHz	+2.25	+1.07	+1.42	+0.50
no-FIR, drv 60 G + MRM 40 GHz	+1.95	+0.77	+1.07	+0.16

Preferred baseline: no-FIR 60 GHz driver + TIA B (single post-cursor tap retained as an option pending measured MRM EO response). The FIR buys 0 dB of ISI on this channel and costs 0.27 dB of slice-DCD jitter.

6. Margin sensitivity

GEN2 margin sensitivity — one lever at a time, all else at baseline



Each bar starts at the +1.34 dB baseline (the waterfall margin) and ends at the margin that results if that one lever moves, all else unchanged; the label is the resulting margin. All levers stay right of 0 dB — no single slip breaks the link.

PD capacitance (30–40 fF assumed, unmeasured) is not a margin lever but a **buildability gate**: 60 fF strands the 4.0 μ A TIA noise line.

7. Status and confidence

Item	Status	Confidence
GEN1 design-point TIA (i_n 3.17 μ A, 29.3 GHz), R = 0.876 A/W	Measured	High
f^2 -dominated noise scaling law	Measured + fit	High

Item	Status	Confidence
GD-ripple failure mode ($12.5 \text{ ps} \rightarrow h_{-1} \approx 0.48$)	Measured	High
TIA noise ceiling $\leq 4.0 \text{ }\mu\text{A}$	Derived	Medium
TIA bandwidth window 50–64 GHz	Derived	Medium
GD ripple $\leq 3 \text{ ps}$ / peaking $\leq 1 \text{ dB}$	Derived	Medium
Microbump $\leq 25 \text{ fF}$ (at assumed $R_{\text{eff}} = 50 \text{ }\Omega$)	Derived	Medium
$B_n = 1.5 \times f_{3\text{dB}}$	Convention	Medium
End reflectance $\leq -24 \text{ dB}$	Required	Medium
PD capacitance 30–40 fF	Assumed	Low
MRM EO bandwidth 40–80 GHz	Assumed	Low
MRM peaking-free response	Assumed	Low
RJ 141 fs rms	Assumed	Low
+1.34 dB typical margin	Model result	Medium

8. Key derived requirements

Requirement	Value
GEN2 TIA class	§4 table above
Noise integration bandwidth	$B_n = 1.5 \times f_{3\text{dB}}$, both generations
End reflectance	$\leq -24 \text{ dB}$ both ends, both generations
Tx 20–80 % transition	$\leq 5.6 \text{ ps}$ (0.60 UI) hard / 4.2 ps (0.45 UI) target
Microbump	$\leq 25 \text{ fF}$, $\leq 30 \text{ pH}$
Jitter	$\text{RJ} \leq 141 \text{ fs rms}$, $\text{DJ} \leq 1.32 \text{ ps}$ (TJ 3.30 ps)
TDEC	$\leq 1.8 \text{ dB}$

9. Kill-or-confirm experiments

1. PD capacitance + MRM EO response (magnitude + phase) at 106 GBd bandwidths.
2. Candidate-TIA magnitude, phase, and noise spectrum to $\geq 90 \text{ GHz}$.
3. Clock/CDR RJ characterization vs the 141 fs target (169 fs fallback: -0.19 dB).
4. Full measured-chain pulse-response closure (driver $\rightarrow$ bump $\rightarrow$ MRM $\rightarrow$ fiber $\rightarrow$ PD $\rightarrow$ bump $\rightarrow$ TIA).